

Student Attrition Machine Learning

AI agent work log | Curated human prompts and design-relevant agent outcomes

Source conversation: Sprint 2 implementation session (6-19 August 2026)

Authoritative code: Machine Learning Model.ipynb

Prepared: 20 August 2026 | Chronological order (oldest to newest)

Models used: a mixture of Composer 2.5, Grok 4.6, Opus 5.0, and ChatGPT Sol 5.6.

This document is not a full chat transcript. Operational noise (authentication, tooling, pedagogy Q&A, interrupted retries) is omitted. Each entry below is a professionally rewritten formal prompt followed by the design-relevant agent outcome that affected the approved model, data pipeline, or Spec-Driven Development (SDD) documentation. Formal SDD prompts (master + Phase 1-3) remain in the Markdown prompt files; this log records the human-agent refinements that produced amendments A-1 to A-6 and related platform workarounds.

How to read each entry. Format descends within every entry: Formal prompt (human instruction, rewritten for clarity) → Design-relevant outcome (what the agent produced or resolved) → Recorded result (amendment, artefact, or decision).

Entry 01 | 2026-08-06 | Project kickoff

Establish implementation authority and knowledge inventory

Formal prompt. Authenticate to Databricks and implement the Sprint 2 student-attrition machine-learning workflow in the Databricks workspace, using the supplied Spec-Driven Development documentation, synthetic-data generator, physical solution design, and Product Backlog context as the controlling knowledge set. Read and inventory those materials before writing code.

Design-relevant outcome. The agent established the authority order used for the rest of the project: controlling SDD Markdown files first; human-approved phase prompts second; project-context documents third; generator/schema evidence fourth. Implementation was constrained to one Spark ML Random Forest, one student-level target (`is_twelve_month_student_attrition`), three human-gated phases, and the approved output table `workspace.student_aggregate.student_attrition_risk_prediction`.

Recorded result: Implementation charter fixed; formal master + Phase 1 prompts become the operating contract.

Entry 02 | 2026-08-06 | Phase 1 refinement → A-1, A-3

Resolve synthetic key collisions and expand usable STAR attributes

Formal prompt. Prioritise the duplicate business-key defect in the synthetic-data generator. Implement the recommended key-collision fix, then ensure usable STAR-schema attributes are applied in modelling, including `course_admission_load_category`. Resolve these issues before Phase 2 begins.

Design-relevant outcome. Profiling showed Spark lpad truncation collapsed dimension keys above 9,999, so unit and course-offering joins would have multiplied the fact table by millions of rows. The generator key width was raised from 4 to 12 characters with uniqueness assertions. Independently, review directed inclusion of fully populated, prediction-time-valid attributes not already represented, taking the approved set from 19 to 21 features by adding `course_admission_load_category` and `commencing_continuing_period`.

Recorded result: Amendments A-1 and A-3. Human confirmed regeneration of all Delta tables.

Entry 03 | 2026-08-06 | Phase 1 refinement → A-2

Confirm canonical field-of-education ownership

Formal prompt. Provide a clear rationale for excluding Study Area A/B field-of-education attributes so that the course dimension remains the single canonical owner of that concept, before any Phase 1 acceptance decision.

Design-relevant outcome. Column-by-column profiling of all 12 generated tables confirmed that study-area and unit FoE columns duplicate a concept already owned by the course dimension, add no independent information beyond chance agreement rates, and (for unit) risk join inflation when keys are non-unique. Exhaustive coverage also confirmed

38 further fact columns are entirely NULL. Canonical ownership was retained over exhaustive column inclusion.

Recorded result: Amendment A-2. Final approved feature set locked at 21 source features.

Entry 04 | 2026-08-06 | Phase 1 gate
Approve Phase 1 and authorise Phase 2

Formal prompt. Phase 1 is approved on the evidence that the notebook executes successfully end-to-end. Record the human review in the SDD documents, mark Phase 1 tasks complete, and begin Phase 2 using the approved Phase 2 prompt.

Design-relevant outcome. Tasks T1-T8 were marked complete. Phase 1 evidence recorded: 973,770 one-row-per-student modelling rows; 5.51% positive class; joins preserved grain; limitations L-1 to L-3 documented (FoE decoupling, flat features, uncalibrated weighted probabilities). Serverless rejection of cache() was recorded as a standing constraint for later phases.

Recorded result: Phase 1 closed 2026-08-06. Phase 2 implementation authorised.

Entry 05 | 2026-08-06 | Clarification (no change)
Retain the approved 0.50 decision threshold

Formal prompt. Clarify whether the fixed 0.50 probability threshold remains appropriate for the current Sprint 2 scope before any threshold redesign is considered.

Design-relevant outcome. The agent confirmed that 0.50 is the approved specification threshold, applied to the unrounded positive-class probability so that evaluation metrics and the Phase 3 risk flag remain consistent. Threshold redesign, cost-sensitive cut-point search, and probability calibration were deferred as out of scope for Sprint 2.

Recorded result: No amendment. Threshold remains 0.50.

Entry 06 | 2026-08-06 | Clarification (no change)
Defer probability calibration

Formal prompt. Explain the role of probability calibration (for example CalibratedClassifierCV) relative to Random Forest vote fractions, and confirm whether any such change should be implemented now.

Design-relevant outcome. Class weighting recentres scores near 0.50 so the approved threshold is usable, but the stored percentage is then a ranking score rather than a calibrated likelihood (limitation L-3). Calibration could improve probability quality but would change model behaviour and require a material-refinement review. The human directed that calibration not be implemented in Sprint 2.

Recorded result: No amendment. L-3 retained as the governing interpretation of stored percentages.

Entry 07 | 2026-08-06 | Phase 2 gate → A-4
Approve Phase 2 and authorise Phase 3; record hash-based split

Formal prompt. Phase 2 is approved in its current form. Mark Phase 2 tasks complete, update controlling documentation as required, and proceed to Phase 3 inference and Delta persistence.

Design-relevant outcome. During Phase 2, randomSplit was replaced by a deterministic hash of student_deidentified_hash into 100 buckets (70/15/15) because serverless compute forbids the cache() that would make randomSplit stable across actions. That material refinement was recorded as amendment A-4. Tasks T9-T21 were marked complete. Persistence was authorised conditional on Phase 3 pre-write validation.

Recorded result: Amendment A-4. Phase 2 closed; Phase 3 authorised.

Entry 08 | 2026-08-06 | Phase 3 design judgement
Confirm persisted schema and post-write verification ordering

Formal prompt. Confirm two Phase 3 design judgements: (1) whether the unrounded probability may remain an intermediate that is validated but not persisted under the six-column approved schema; and (2) whether percentage distribution statistics may be computed from the written Delta table rather than from a second scoring pass.

Design-relevant outcome. Widening the persisted schema would require a formal amendment; the unrounded probability is therefore carried only in the transient staging table and dropped at publication. Post-publish checks read the approved Delta table so they verify what the application will retrieve. Combined with the serverless cache ban, the scored result is materialised once to `student_attrition_risk_prediction__scoring_staging`, validated, published, then dropped.

Recorded result: No schema amendment. Staging-then-publish architecture confirmed.

Entry 09 | 2026-08-06 | Platform workaround **Enable MLflow Spark model logging on serverless**

Formal prompt. Resolve the MLflow failure stating that a Unity Catalog volume path must be provided to save, log, or load Spark ML models on Databricks shared or serverless clusters.

Design-relevant outcome. Serverless compute cannot stage Spark ML artefacts on local disk. The notebook creates `workspace.student_aggregate.mlflow_tmp` idempotently and passes it as `dfs_tmpdir` to `mlflow.spark.log_model`. A pre-training write/delete smoke test was added because the forest fit is expensive and this failure previously surfaced only after paying for training.

Recorded result: Routine correction (no behavioural amendment). MLflow logging unblocked.

Entry 10 | 2026-08-06 | Platform workaround **Correct one-hot feature-importance slot accounting**

Formal prompt. Resolve the Phase 2 assertion that feature-importance slot counts (260) do not match the fitted feature-vector width (274).

Design-relevant outcome. The mismatch came from reading encoder categorySizes without the extra slot created by invalid/unseen handling. Slot widths were taken from the fitted pipeline output schema via `AttributeGroup`, then summed back to the 21 approved source features for reporting only. No feature was added, removed, or reweighted.

Recorded result: Routine correction. Importance reporting aligned with the assembled vector.

Entry 11 | 2026-08-06 | Platform workaround **Contain Spark Connect model-cache overflow**

Formal prompt. Resolve repeated `ML_CACHE_SIZE_OVERFLOW_EXCEPTION` failures on Databricks free-edition Serverless when refitting the Spark ML model within the same session, and identify the practical execution guidance for that environment.

Design-relevant outcome. Serverless caps a single model at 100 MB and the session model cache at 1 GB. Spark Connect retains fitted models while Python references survive, including traceback references. The notebook was updated to release prior fitted objects before training and to assert the fitted tree count immediately after fit. Operators must restart the Python session for clean re-runs. These constraints later motivated parameter reduction (A-5) and removal of fitted preprocessing estimators (A-6).

Recorded result: Routine session-reset handling. Platform limits documented for A-5 and A-6.

Entry 12 | 2026-08-06 | Phase 3 gate **Approve Phase 3 and close Sprint 2 machine-learning implementation**

Formal prompt. Phase 3 is approved. Update notebook documentation cells and Spec-Driven Development documents to reflect the approved outcome. Do not change executable model behaviour.

Design-relevant outcome. Tasks T22-T32 were marked complete. The approved prediction table was confirmed queryable with one row per student_deidentified_hash. Documentation recorded staging-then-publish, six-column persistence, inference over the full modelling population, and continued application of limitations L-1 to L-3. Parent Product Backlog stories remain the project-level acceptance gate.

Recorded result: Phase 3 closed 2026-08-06. Sprint 2 ML implementation complete at that gate.

Entry 13 | 2026-08-07 | Material refinement → A-5
Reduce Random Forest size for free-edition memory limits

Formal prompt. Update plan.md to the Random Forest configuration numTrees=100, maxDepth=7, minInstancesPerNode=20, featureSubsetStrategy="sqrt", seed=RANDOM_SEED, and document that the reduction is required for computational reasons so the fitted model fits within Databricks free-edition Serverless memory limits. The corresponding notebook constants were updated by the human reviewer.

Design-relevant outcome. The original 300/10/5 ensemble repeatedly approached the 100 MB per-model and 1 GB session-cache caps. The approved reduction preserves the Spark ML Random Forest workflow, feature set, split, threshold, and output schema while trading some ensemble capacity for reliable free-tier execution. Recall remains the primary metric.

Recorded result: Amendment A-5 (approved 2026-08-07).

Entry 14 | 2026-08-16 to 2026-08-19 | Material refinement → A-6
Reconcile SDD documentation to stateless preprocessing architecture

Formal prompt. Using the newest Machine Learning Model.ipynb as the authoritative implementation, update the Spec-Driven Development documentation so it accurately describes training-derived numeric medians, training-derived categorical levels, deterministic manual one-hot encoding with an explicit unseen/NULL bucket, VectorAssembler used only as a Transformer, and RandomForestClassifier fitted directly to the prepared training vector. Record the change as amendment A-6 after A-5. Do not rewrite history as though the original fitted Spark ML Pipeline never existed. Do not modify executable code.

Design-relevant outcome. After human approval of the proposed change summary, documentation was updated across plan.md, specification.md (US-7), tasks.md, Phase 2 and Phase 3 prompts, and the master prompt. Active requirements for Imputer, StringIndexer, OneHotEncoder, fitted preprocessing Pipeline, and attrition_pipeline were removed from current-design sections. A-6 records the Spark Connect fitted-graph workaround while preserving the train-only statistical contract for validation, test, and inference.

Recorded result: Amendment A-6. SDD set aligned to current notebook; code unchanged.

Entry 15 | 2026-08-19 | Implementation documentation
Produce prediction-output and implementation-traceability documentation

Formal prompt. Produce two PDF documents from the current approved implementation: (1) prediction-output documentation describing the output table, schema, derivation rules, validation, and retrieval contract; and (2) an implementation-traceability document covering the most important development prompts, model assumptions, feature exclusions, and design decisions, suitable for placement in the GitHub repository.

Design-relevant outcome. Two PDFs were written to document the prediction output contract and the implementation traceability. Live per-row scores remain in the Delta table; the predictions PDF documents the output contract because the notebook export contains no executed score dumps and warehouse access was unavailable at generation time.

Recorded result: Implementation-traceability documentation created for the GitHub repository.

Entry 16 | 2026-08-19 | This document
Curate the professional AI-agent work log

Formal prompt. In addition to the formal master and phase prompts, identify the other development prompts that materially shaped the implementation. Produce a third PDF that presents those prompts in professional language, each

followed by the design-relevant agent outcome, in chronological order, excluding off-topic or low-value conversation.

Design-relevant outcome. This work log was produced as Student_Attrition_ML_Agent_Work_Log.pdf. It retains only kickoff, Phase 1-3 gates, amendments A-1 to A-6, platform workarounds that changed architecture, and implementation-traceability documentation prompts. Authentication, tooling, and pedagogy Q&A were excluded.

Recorded result: Work-log PDF for GitHub.

Summary of design-changing outcomes

ID / event	Effect
A-1	Feature set 19 → 21
A-2	Canonical FoE ownership retained
A-3	Synthetic key integrity restored; tables regenerated
A-4	Hash-based 70/15/15 split
A-5	RF parameters 100 / 7 / 20 for Serverless memory
A-6	Stateless preprocessing; direct RF fit
MLflow UC volume	Serverless Spark ML artefact logging
Staging table	Score once; validate; publish six columns
L-1 to L-3	Govern interpretation of metrics and stored percentages

Companion documentation: the prediction-output PDF and Student_Attrition_ML_Implementation_Traceability.pdf. Formal prompt files remain in the SDD package: AI_AGENT_MASTER_PROMPT.md, PHASE_1_FACT_CENTRED_FEATURE_ASSEMBLY.md, PHASE_2_MODEL_TRAINING_EVALUATION.md, PHASE_3_INFERENCE_DELTA_PERSISTENCE.md.

Human review. This document has been human reviewed by a member of the team and is approved.